

Disease and Illness Data Dictionary from IDSP

Managing public health surveillance data accurately is challenging because disease reports originate from diverse hospitals, district health centers, and state surveillance units under the Integrated Disease Surveillance Programme (IDSP). As reporting systems update, local health staff input data manually, or clinical terminologies change, disease names are frequently entered inconsistently. If these variations are not properly accounted for, they result in duplicate disease entries, fragmented outbreak tracking, and an inaccurate representation of public health statistics.

While surveillance portals provide disease outbreak reports, raw data feeds contain significant noise such as random character spaces, spelling errors, colloquial disease terms, and inconsistent co-infection reporting. At Dataful, we realised this challenge and have put in significant efforts to clean up, reconcile, and provide accurate, standardized data.

Standardization simply means unifying variations of disease names into clear, consistent master terms. Without this cleanup, analytical software and health tracking dashboards make critical errors such as splitting a single disease outbreak across multiple entries, distorting case counts, or failing to aggregate co-infections accurately.

Issues in IDSP Records

Dataful's cleaning strategy systematically resolves these issues across four primary categories:

- *Typos, Formatting, and Text Variations*: Corrects spelling errors, broken character spacing, shorthand abbreviations, and minor text additions into uniform master entries.

Eg: Raw health logs split single words into broken strings such as "*Acute Gastroente* *ritis*", "*Acute Gastroenter itis*", "*Acute Gastroenteri tis*", and "*Leptospiro sis*".

Standardizing all of these unifies case counts under clean medical terms like Acute Gastroenteritis and Leptospirosis. Similarly, local abbreviations like "*HFMD*" and "*PUO*" are expanded to Hand, Foot and Mouth Disease and Fever of Unknown Origin.

- *Linguistic & Clinical Spelling Conventions*: Reconciles regional spelling variations (such as British vs. American English) and clinical terminology shifts across different health surveillance units into a single reference standard.

Eg: Outbreak logs alternate between British ("*Acute Diarrhoeal Disease*") and American ("*Acute Diarrheal Disease*") spellings, or alternate terms like "*Acute Encephalitis Syndrome*" and "*Acute Encephalitic Syndrome*". The dictionary maps all of these variations to uniform master terms (Acute Diarrheal Disease and Acute Encephalitic Syndrome), ensuring all the reported cases aggregate into a single continuous time series.

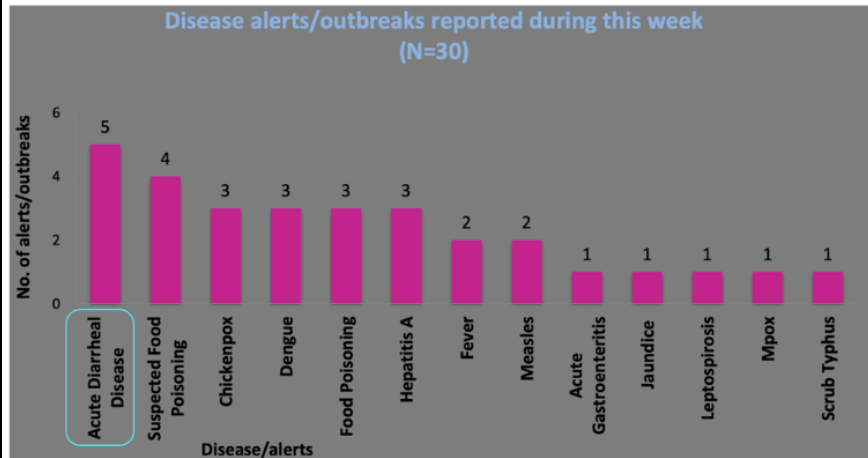
- *Ordering of Co-Infections and Dual Diagnoses:* Standardizes multi-disease outbreak reports where two pathogens or illnesses are reported together, preventing identical co-infection events from being split based on word order in raw filings.

Eg: Raw reports list identical co-infections in different sequences, such as "*Chikungunya & Dengue*", "*Dengue & Chikungunya*", or "*Dengue and Chikungunya*". Standardizing word order and conjunctions unifies these under a single master entry (Chikungunya and Dengue). The same approach is applied to entries like "*Measles and Chickenpox*" (standardized to Chickenpox and Measles) and "*Scrub Typhus and Leptospirosis*" (standardized to Leptospirosis and Scrub Typhus).

- *Strain Normalization and Nomenclature Shifts Across Reports:* Unifies variations in how specific strains, species, or clinical sub-types are recorded across different health facilities and monthly surveillance releases.

Eg: Raw registries capture mixed strain infections under multiple text formats—including "*Malaria (P. vivax & P.falciparum)*", "*Malaria (P.falciparum & P.vivax)*", "*Malaria (P.falciparum and P.vivax)*", and "*Malaria (Plasmodium Falciparum)*". Standardizing these variations into clean, uniform nomenclature (such as Malaria (P.falciparum & P.vivax), Malaria (P.falciparum), and Malaria (P.vivax)) prevents multi-year time-series analysis from breaking and ensures strain-specific tracking remains precise.

Reporting Inconsistencies within IDSP's Weekly Report: Official weekly reports published by the IDSP frequently feature inconsistent naming conventions for the exact same disease even within the same release or across consecutive reporting periods. In just the IDSP's weekly disease report for the [43rd week of 2025](#), one can observe how different spellings and text entries have been used for the same disease. Standardizing these variations into a unified master entry prevents multi-year time-series analysis from breaking and ensures epidemiological metrics—such as outbreak severity and case counts—are tracked accurately over time.



CT/NAR/2025/43/1929	Chhattisgarh	Narayanpur	Acute Diarrhoeal Disease	25	5	21-10-2025	24-10-2025	Under Surveillance
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